United States Patent	12408957
Kind Code	B2
Date of Patent	September 09, 2025
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Orthopaedic implant system for compression-distraction of joints

Abstract

An orthopaedic implant system includes an orthopaedic implant having a plurality of fixation openings and a plug opening defining a plug opening diameter that is greater than a diameter of any of the fixation openings. The plurality of fixation openings include at least one first fixation opening configured to accept a fixation element inserted in a first bone and at least one second fixation opening configured to accept a fixation element inserted in a second bone. The plug opening is configured to accept a plug inserted in a joint space between the first bone and the second bone to cause compression distraction of the first bone and the second bone.

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Appl. No.: **17/945944**

Filed: **September 15, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230088758 A1	Mar. 23, 2023

Related U.S. Application Data

us-provisional-application US 63246552 20210921

Publication Classification

Int. Cl.: **A61B17/80** (20060101); **A61B17/02** (20060101); **A61B17/66** (20060101); **A61B17/68** (20060101); **A61B17/72** (20060101); **A61F2/30** (20060101); **A61F2/42** (20060101)

U.S. Cl.:

CPC **A61B17/8004** (20130101); A61B17/025 (20130101); A61B17/66 (20130101); A61B17/68 (20130101); A61B17/686 (20130101); A61B17/7216 (20130101); A61B17/7225 (20130101); A61B17/8014 (20130101); A61B17/8061 (20130101); A61F2002/30622 (20130101); A61F2/4225 (20130101); A61F2002/4228 (20130101); A61F2002/4233 (20130101); A61F2002/4238 (20130101); A61F2/4241 (20130101); A61F2002/4243 (20130101); A61F2002/4251 (20130101); A61F2/4261 (20130101)

Field of Classification Search

CPC: A61B (17/8004); A61B (17/80); A61B (17/8014); A61B (17/8061); A61B (17/808); A61B (17/66); A61B (17/68); A61B (17/025); A61B (17/7216); A61B (17/7225); A61B (17/686); A61B (2017/681); A61F (2/42); A61F (2/4225); A61F (2/4241); A61F (2/4261); A61F (2002/4228); A61F (2002/4233); A61F (2002/4238); A61F (2002/4243); A61F (2002/4251); A61F (2002/30622)

USPC: 606/282; 606/86R; 606/87; 606/90; 606/105; 606/86B; 606/902; 606/906; 606/915; 606/280; 606/70; 606/281; 606/286; 606/298; 606/300

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This is a non-provisional application based upon U.S. provisional patent application Ser. No. 63/246,552, entitled “ORTHOPAEDIC IMPLANT SYSTEM FOR COMPRESSION-DISTRACTION OF JOINTS”, filed Sep. 21, 2021, which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to medical implants, and, more particularly, to orthopaedic implants.

2. Description of the Related Art

(3) Many types of orthopaedic surgeries and associated implants are known. One type of known orthopaedic surgery is a fusion surgery, in which two or more bones are fused together using an

orthopaedic implant, such as a plate. When a plate is used to perform a fusion, the plate must cross over the natural joint space between the two or more bones.
(4) There are several issues that can detrimentally affect the success of a fusion surgery. In some instances, the fusion site may be prepared in a manner that excessively shortens the overall length of the involved bones, which can lead to unnatural biomechanics at the fusion site. It is also possible that fusion between the bones may not occur due to a lack of compression at the fusion site.
(5) What is needed in the art is a way to address some of the issues with known fusion surgeries and associated orthopaedic implants.
SUMMARY OF THE INVENTION
(6) The present invention provides an orthopaedic implant system including an orthopaedic implant with a plug opening formed therein that is a largest opening of the orthopaedic implant and is sized and configured to accept a plug that can cause distraction of bones and compression of a graft against bone ends when inserted between bones.
(7) In some exemplary embodiments provided according to the present invention, an orthopaedic implant system includes an orthopaedic implant having a plurality of fixation openings and a plug opening defining a plug opening diameter that is greater than a diameter of any of the fixation openings. The plurality of fixation openings include at least one first fixation opening configured to accept a fixation element inserted in a first bone and at least one second fixation opening configured to accept a fixation element inserted in a second bone. The plug opening is configured to accept a plug inserted in a joint space between the first bone and the second bone to cause compression distraction of the first bone and the second bone.
(8) In some exemplary embodiments provided according to the present invention, a method of implanting an orthopaedic implant is provided. The method includes: positioning the orthopaedic implant so at least one first fixation opening of the orthopaedic implant is aligned with a first bone and at least one second fixation opening of the orthopaedic implant is aligned with a second bone; inserting a first fixation element into the first bone through the at least one first fixation opening; and inserting a second fixation element into the second bone through the at least one second fixation opening. A plug opening of the orthopaedic implant is aligned with a joint space between the first bone and the second bone after inserting the first fixation element into the first bone and after inserting the second fixation element into the second bone. The plug opening defines a plug opening diameter that is greater than a diameter of any of the fixation openings and is configured to accept a plug inserted in the joint space to cause compression distraction of the first bone and the second bone.
(9) A possible advantage that may be realized by exemplary embodiments provided according to the present invention is that the plug opening can accept a plug that compresses a graft and/or the bones at the fusion site to maintain the length integrity of the anatomical structures.
(10) Another possible advantage that may be realized by exemplary embodiments provided according to the present invention is that the plug can provide compression across the joint line to support fusion and help keep the joint in a distracted position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:
(2) FIG. **1** is a top view of an exemplary embodiment of an orthopaedic implant provided according

to the present invention;
(3) FIG. **2** is a side view of an exemplary embodiment of a plug provided according to the present invention;
(4) FIG. **3** is a top view of an exemplary embodiment of an orthopaedic implant system including the orthopaedic implant of FIG. **1** as well as the plug of FIG. **2** inserted in a plug opening of the orthopaedic implant;
(5) FIG. **4** is a side view of the orthopaedic implant system of FIG. **3**;
(6) FIG. **5**A is a cross-sectional view of an exemplary embodiment of an orthopaedic implant system that is used to fuse a first bone to a second bone including an orthopaedic implant including a plug opening that holds a plug inserted in a joint space between the bones;
(7) FIG. **5**B is a section view taken along line A-A in FIG. **5**A;
(8) FIG. **5**C is a top view of the orthopaedic implant of FIG. **5**A;
(9) FIG. **6**A is a side view of an exemplary embodiment of a plug provided according to the present invention for use in an orthopaedic implant system;
(10) FIG. **6**B is a top view of the plug of FIG. **6**A;
(11) FIG. **6**C is a bottom view of the plug of FIGS. **6**A and **6**B;
(12) FIG. **7** is a perspective view of another exemplary embodiment of an orthopaedic implant provided according to the present invention;
(13) FIG. **8** is a top view of another exemplary embodiment of an orthopaedic implant system provided according to the present invention that includes an orthopaedic implant with multiple plug openings and a plug disposed in each of the plug openings; and
(14) FIG. **9** is a flow chart illustrating an exemplary embodiment of a method of implanting an orthopaedic implant according to the present invention.
(15) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the invention and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE INVENTION

(16) Referring now to the drawings, and more particularly to FIGS. **1-4**, there is shown an exemplary embodiment of an orthopaedic implant system **300** (first illustrated in FIG. **3**) which generally includes an orthopaedic implant **100** (first illustrated in FIG. **1**) and, in some embodiments, a plug **200** (first illustrated in FIG. **2**) that is coupled to the orthopaedic implant **100**. The orthopaedic implant **100** is shaped and sized to be used for fusing the cuneiform to the first metatarsal, but it should be appreciated that the shape and dimensions of the orthopaedic implant **100** can be adjusted for use in other joint areas, as will be appreciated further herein. The orthopaedic implant **100** includes a plurality of fixation openings **101** formed therein that can accept a fixation element, such as a bone screw, to fixate the orthopaedic implant **100** to the bones of the joint. At least one of the fixation openings **101** may be referred to as a first fixation opening that is configured to accept a fixation element inserted in a first bone and at least one of the other fixation openings **101** may be referred to as a second fixation opening that is configured to accept a fixation element inserted in a second bone, as will be described further herein. The fixation elements may be variable angle or fixed angle fixation elements, which are both known in the art. The orthopaedic implant **100** may be formed of any suitable biocompatible material, including but not limited to: metals such as titanium, cobalt-chrome, stainless steel, and/or alloys thereof; or polymers such as ultra-high molecular weight polyethylene (UHMWPE), polyarylether ketones (PAEK) such as polyether ether ketone (PEEK), and/or blends thereof. The biocompatible material may be partially resorbable, fully resorbable, or minimally resorbable (less than 5% resorption within one year of implantation).
(17) The orthopaedic implant **100** also includes a single plug opening **102** that is a largest opening of the orthopaedic implant **100**. The plug opening **102** may, for example, define a plug opening diameter POD that is a relatively high percentage of a greatest width GW of the orthopaedic

implant **100**, e.g., the plug opening diameter POD may be greater than 50% of the greatest width GW, such as 50-90% of the greatest width GW. In addition, or alternatively, the plug opening **102** may define a plug opening diameter POD that is several times a fixation opening diameter FOD of the fixation openings **101**. For example, the plug opening diameter POD may be at least double the fixation opening diameter FOD or even at least triple the fixation opening diameter FOD. In other words, the plug opening diameter POD is greater than the fixation opening diameter FOD of any of the fixation openings **101**. While the openings **101**, **102** are illustrated as being circular, in some embodiments the openings **101**, **102** are non-circular or even non-round; in such embodiments, the plug opening diameter POD may correspond to a plug opening width that is a largest dimension of the plug opening **102**. Further, it should be appreciated that while the orthopaedic implant **100** is illustrated with a single plug opening **102**, in some embodiments the orthopaedic implant **100** includes multiple plug openings, e.g., if the orthopaedic implant spans across more than two bones such as a first bone, a second bone, and a third bone, that are considerably larger than any other openings of the orthopaedic implant **100**. In embodiments with multiple plug openings, the diameter of each plug opening may be less than 50% of the greatest width of the orthopaedic implant if necessary, e.g., to prevent the plug openings from overlapping and/or splitting the orthopaedic implant in half.

(18) As best illustrated in FIG. **4**, the orthopaedic implant **100** may have two regions **103**A, **103**B where the openings **101**, **102** are formed that are connected by an offsetting region **104** so the regions **103**A, **103**B are offset from one another. As can be appreciated from FIG. **4**, one of the regions **103**A, which may be referred to as a second region including the at least one second fixation opening **101**, may have a bottom surface **105** that is above a corresponding bottom surface **106** of the other region **103**B, which may be referred to as a first region including the at least one first fixation opening **101**. In this respect, the bottom surface **106** may represent the bottom surface **106** of the orthopaedic implant **100**. Similarly, the second region **103**A may have a top surface **107** that is above a corresponding top surface **108** of the first region **103**B, with the top surface **107** representing the top surface **107** of the orthopaedic implant **100**.

(19) Referring specifically now to FIG. **2**, the plug **200** that is coupled to the orthopaedic implant **100** to form the orthopaedic implant system **300** is illustrated in greater detail and separate from the orthopaedic implant **100** to better illustrate the features of the plug **200**. The plug **200** may include a head **201**, a cylindrical section **202** connected to the head **201**, and a tapered section **203** connected to the cylindrical section **202**. The head **201** may define a head diameter HD that is greater than the plug opening diameter POD to prevent the plug **200** from going through the plug opening **102** when inserted therein, while the cylindrical section **202** and the tapered section **203** may both define respective diameters CD, TD that are less than the plug opening diameter POD. As can be appreciated from FIG. **2**, the tapered diameter TD decreases in a direction from the cylindrical section **202** toward a bottom **204** of the plug **200**. It should be appreciated that, in some embodiments, the portion of the plug **200** referred to herein as the “cylindrical section” **202** may be tapered, in addition to or alternatively to the tapered section **203**. Further, in some embodiments the plug **200** does not include a tapered section. The plug **200** may also include a driver feature **205**, illustrated as a driver opening, formed in a top **206** of the plug **200** that is configured to interact with a driver, such as a hex wrench, to drive the plug **200** into the plug opening **102**, as will be described further herein. The plug **200** may be partially or completely unthreaded or, in some embodiments, partially or completely threaded. In some embodiments, the plug **200** is formed from a polymer.

(20) The plug opening **102** is configured to accept the plug **200** inserted in a joint space between a first bone and a second bone to cause compression distraction of the first bone and the second bone. As illustrated in FIGS. **3-4**, the orthopaedic implant system **300** may be formed by inserting the plug **200** in the plug opening **102** of the orthopaedic implant **100**. The plug **200** may be inserted until the head **201** of the plug **200** abuts against the top surface **107** of the region **103**A, at which

point the bottom **204** of the plug **200** is below the bottom surfaces **105**, **106** of both of the regions **103**A, **103**B of the orthopaedic implant **100**. In this respect, the plug **200** may extend past the orthopaedic implant **100** into a joint space between two adjacent bones to bear on the bones. When fully inserted, the plug **200** is engaged with the orthopaedic implant **100** so the plug **200** is not easily removed from the orthopaedic implant **100**, e.g., by oversizing the plug **200** relative to the plug opening **102** to form an interference fit or by using some other type of engagement such as threads, as will be described further herein.

(21) The plug **200** bearing on the bones may create compression of a graft and/or the graft creates compression against bone ends of the bones while also distracting the bones, leading to compression distraction that can maintain the length integrity of the anatomical structures rather than shortening the length, which is commonly encountered when performing a fusion surgery. In some embodiments, the portion(s) of the plug **200** that extends out of the plug opening **102**, when the plug **200** is fully inserted, defines a protruding length of 2-4 mm or 1-7 mm.

(22) Referring now to FIGS. **5**A-**5**C, another exemplary embodiment of an orthopaedic implant system **500** is illustrated that includes an orthopaedic implant **510** and a plug **520**. The plug **520** may be similar to the previously described plug **200**. The orthopaedic implant **510** has a plurality of fixation openings **511**A, **511**B and a single plug opening **512**. The plug **520** is inserted into the plug opening **512**. As can be appreciated from FIG. **5**C, which illustrates the orthopaedic implant **510** by itself, three of the fixation openings **511**A may be aligned with one another so their centers are located along a longitudinal axis LA of the orthopaedic implant **510** while two of the fixation openings **511**B may be aligned with one another so their centers are located along a width axis WA of the orthopaedic implant **510**. It should be appreciated that the arrangement of the fixation openings **511**A, **511**B may be adjusted, as desired. The orthopaedic implant **510** includes a first region **513**A that includes the fixation openings **511**A and a second region **513**B that includes the plug opening **512** and the fixation openings **511**B. The first region **513**A may be connected to the second region **513**B by an offsetting region **514** so the first region **513**A is lower than the second region **513**B. The offsetting region **514** can create a "step" between the first region **513**A and the second region **513**B, with the step being between, for example, 2-4 mm, depending on the anatomical structures being fused by the orthopaedic implant **510**. It should be appreciated that the length and angle of the offsetting region **514** relative to the regions **513**A, **513**B can adjusted, as desired, to create a step of a desired length, which may be more or less than the previously described 2-4 mm such as 1-8 mm.

(23) As can be seen in FIG. **5**A, the first region **513**A may be fixated to a first bone B**1** by fixation elements extending through the fixation openings **511**A, which may also be referred to as first fixation openings. The second region **513**B, on the other hand, may be fixated to a second bone B**2** by fixation elements extending through the fixation openings **511**B, which may also be referred to as second fixation openings. A joint space JS is established between the first bone B**1** and the second bone B**2**. The orthopaedic implant **510** may be placed so the first region **513**A is properly placed on the first bone B**1** and the second region **513**B is properly placed on the second bone B**2**, with the plug opening **512** being located above, i.e., aligned with, the joint space JS. The regions **513**A, **513**B may be initially fixed with one or more fixation elements, such as bone screws, inserted in the fixation openings **511**A, **511**B and into the bones B**1**, B**2**. The initial fixation can hold the orthopaedic implant **510** in place during subsequent surgical steps.

(24) A plug of tissue may be harvested from the joint space JS and replaced with a graft **530** that may be greater in size than the harvested plug of tissue. The plug of tissue may be harvested through the plug opening **512** of the orthopaedic implant **510**, which may be left open, so the orthopaedic implant **510** does not need to be positioned following plug harvesting. It should thus be appreciated that the harvested plug of tissue may have a diameter (or width) that is smaller than the corresponding diameter (or width) of the plug opening **512**. The graft **530** may also be inserted into the joint space JS through the plug opening **512**. The graft **530** may be, for example, biological

tissue such as an autograft, allograft, etc., a scaffold configured to allow tissue ingrowth, or a combination of both. The graft **530** may have a diameter that is 1-2 mm greater than that of the harvested plug of tissue so the graft **530** can compress against the bones **510**, **520**. After the graft **530** is placed, additional fixation elements can be inserted into the fixation openings **511**A, **511**B and the bones B**1**, B**2** to fully secure the orthopaedic implant **510** to the bones B**1**, B**2** and the plug **520** can be inserted into the plug opening **512** to cause compression distraction of the first bone B**1** and the second bone B**2**. In some embodiments, the plug **520** is inserted into the plug opening **512** prior to inserting additional fixation elements into the fixation openings **511**A, **511**B; in some embodiments, additional fixation elements are inserted into the fixation openings **511**A, **511**B prior to inserting the plug **520** into the plug opening **512**.
(25) The plug **520** extends through the plug opening **512** so a portion of the plug **520** compresses the graft **530**, which maintains the graft **530** in place. As the plug **520** is inserted, a tapered section **523** of the plug **520** can gradually create distance between the bones B**1**, B**2** to distract the bones B**1**, B**2**. A portion of the plug **520**, such as a cylindrical section **522** of the plug **520**, can contact cortical bone after the plug **520** is fully inserted in the plug opening **512** to distract the bones B**1**, B**2** and maintain the length integrity of the anatomical structures by not allowing the bones B**1**, B**2** to migrate toward one another. Thus, the plug **520** can create compression between the graft **530** and the bones B**1**, B**2** and distraction on the bones B**1**, B**2** so the anatomical structures do not shorten following the fusion. It should be appreciated that while the procedure of implanting an orthopaedic implant system **500** is described in the context of FIGS. **5**A-**5**C, a similar procedure can be utilized for other orthopaedic implant systems provided according to the present invention, including but not limited to the previously described orthopaedic implant system **300** and orthopaedic implant systems described further herein.
(26) Referring now to FIGS. **6**A-**6**C, another exemplary embodiment of a plug **600** that can be used in an orthopaedic implant system is illustrated. The plug **600** may have a similar shape to the previously described plugs **200**, **520** and have threads **604** formed on an exterior of the plug **600**, such as a cylindrical section **602** of the plug **600**. In some embodiments, some of the threads **604** also extend to a head **601** of the plug **600**. The plug **600** may also have a tapered section **603** connected to the cylindrical section **602**. The plug **600** may further include a driver feature **605**, such as a hex-shaped opening, that is configured to accept a driver, such as a hex-head wrench, to drive the plug **600**. By threading the plug **600**, the plug **600** can be driven into an opening formed in bone tissue that has a smaller diameter than the plug **600**, i.e., the plug **600** is oversized relative to the opening formed in the bone tissue. The plug **600** being oversized relative to the opening formed in the bone tissue allows the plug **600** to tightly fit against the bone tissue, such as cortical bone tissue, and help compress a graft against bones ends and distract the bones. The threads **604** can also interact with corresponding threads formed in a plug opening of an orthopaedic implant, in which the plug **600** is inserted, to hold the plug **600** in place following implantation so the plug **600** is not, for example, squeezed out from the plug opening by the bones compressed by the plug **600**.
(27) Referring now to FIG. **7**, another exemplary embodiment of an orthopaedic implant **700** that may be used in an orthopaedic implant system with a plug is illustrated. The illustrated shape of the orthopaedic implant **700** may be configured, for example, to perform a talar-navicular fusion. As illustrated, the orthopaedic implant **700** has a plurality of fixation openings **701** and a single plug opening **702** formed therein. The plug opening **702** may be considerably larger than any of the fixation openings **701**, as previously described. The plug opening **702** may be placed so it overlaps the geometric center of the orthopaedic implant **700**. The fixation openings **701** may be placed around the plug opening **702**, as illustrated, to fixate the orthopaedic implant **700** to bones. It should thus be appreciated that the orthopaedic implants provided according to the present invention can be formed with many different general shapes and arrangements of openings depending on the anatomical location where the orthopaedic implant is being used, e.g., at various locations in the foot, hand, etc. It should be further appreciated that the orthopaedic implants

provided according to the present invention may be shaped and configured for use to span one joint between two bones or multiple joints, such as two joints, between more than two bones.

(28) As previously described, orthopaedic implants provided according to the present invention may include multiple plug openings that are each configured to accept a respective plug. Referring now to FIG. **8**, an exemplary embodiment of an orthopaedic implant system including an orthopaedic implant **800** provided according to the present invention is illustrated that includes a plurality of fixation openings **801**A, **801**B and a plurality of plug openings **802**A, **802**B, **802**C. The fixation openings **801**A, which may be referred to as first fixation openings, are configured to accept a fixation element inserted in a first bone B**1**, illustrated as a talus, and the fixation openings **801**B, which may be referred to as second fixation openings, are configured to accept a fixation element inserted in a second bone B**2**, illustrated as a navicular. Each of the fixation openings **801**A, **801**B may define a fixation opening diameter FOD that is the same. The plug openings **802**A, **802**B, **802**C, on the other hand, each define a plug opening diameter POD that is greater than the fixation opening diameter FOD of any of the fixation openings **801**A, **801**B. Each of the plug openings **802**A, **802**B, **802**C is configured to accept a respective plug **803**A, **803**B, **803**C inserted in a joint space JS between the first bone B**1** and the second bone B**2**. When a plug **803**A, **803**B, **803**C is inserted in each of the plug openings **802**A, **802**B, **802**C, the inserted plugs **803**A, **803**B, **803**C are positioned in the joint space JS and can cause compression distraction of the two bones B**1**, B**2**. It should thus be appreciated that the orthopaedic implant system of FIG. **8** may be utilized in fusion surgeries where a large joint space is involved to provide sufficient and even compression across the joint space. Alternatively, one or more of the plug openings **802**A, **802**B, **802**C may be configured to accept a respective plug **803**A, **803**B, **803**C inserted in a joint space between, for example the second bone B**2** and a third bone, so the orthopaedic implant **800** fuses together three (or more) bones rather than two bones as illustrated.

(29) Referring now to FIG. **9**, an exemplary embodiment of a method **900** of implanting an orthopaedic implant **100**, **510**, **700**, **800** provided according to the present invention is illustrated. The method **900** includes positioning **901** the orthopaedic implant **100**, **510**, **700**, **800** so at least one first fixation opening **101**, **511**A, **701**, **801**A is aligned with a first bone B**1** and at least one second fixation opening **101**, **511**B, **701**, **801**B is aligned with a second bone B**2**. The method **900** further includes inserting **902** a first fixation element into the first bone B**1** through the at least one first fixation opening **101**, **511**A, **701**, **801**A and inserting **903** a second fixation element into the second bone B**2** through the at least one second fixation opening **101**, **511**B, **701**, **801**B. A plug opening **102**, **512**, **702**, **802**A, **802**B, **802**C of the orthopaedic implant **100**, **510**, **700**, **800** is aligned with a joint space JS between the first bone B**1** and the second bone B**2** after inserting **902** the first fixation element into the first bone B**1** and after inserting **903** the second fixation element into the second bone B**2**. Inserting **902**, **903** the fixation elements may be performed after the plug opening **102**, **512**, **702**, **802**A, **802**B, **802**C is already aligned with the joint space JS, with the inserted fixation elements providing initial fixation to maintain the alignment. The plug opening **102**, **512**, **702**, **802**A, **802**B, **802**C defines a plug opening diameter POD that is greater than a diameter FOD of any of the fixation openings **101**, **511**A, **511**B, **701**, **801**A, **801**B and is configured to accept a plug **200**, **520**, **600**, **803**A, **803**B, **803**C inserted in the joint space JS to cause compression distraction of the first bone B**1** and the second bone B**2**. The method **900** may further include inserting **904** a plug **200**, **520**, **600**, **803**A, **803**B, **803**C into the plug opening **102**, **512**, **702**, **802**A, **802**B, **802**C to cause compression distraction of the first bone B**1** and the second bone B**2**, as previously described. In some embodiments, the method **900** also includes harvesting **905** a plug of tissue from the joint space and inserting **906** a graft **530** into the joint space JS after harvesting **905** the plug of tissue. Harvesting **905** the plug of tissue and inserting **906** the graft **530** may both be performed before inserting the plug **200**, **520**, **600**, **803**A, **803**B, **803**C into the plug opening **102**, **512**, **702**, **802**A, **802**B, **802**C. In some embodiments, the plug of tissue is harvested **905** from the joint space JS and/or the graft **530** is inserted into the joint space JS through the plug opening **102**,

512, **702**, **802**A, **802**B, **802**C without the plug **200**, **520**, **600**, **803**A, **803**B, **803**C inserted therein.
(30) While this invention has been described with respect to at least one embodiment, the present invention can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims.

Claims

1. An orthopaedic implant system, comprising: an orthopaedic implant comprising: a first fixation opening through a first portion of the orthopaedic implant; a second fixation opening through a second portion of the orthopaedic implant; and a plug opening through the orthopaedic implant, the plug opening being positioned between the first fixation opening and the second fixation opening; wherein: a diameter of the plug opening is greater than a diameter of the first fixation opening; the diameter of the plug opening is greater than a diameter of the second fixation opening; the first fixation opening is configured to receive a first fixation element that is configured to be inserted into a first bone; the second fixation opening is configured to receive a second fixation element that is configured to be inserted into a second bone; the plug opening is configured to accept a plug that is configured to be inserted into a joint space between the first bone and the second bone to cause compression distraction of the first bone and the second bone.
2. The orthopaedic implant system of claim 1, further comprising a plug disposed in the plug opening, the plug being configured to cause compression distraction of the first bone and the second bone when inserted in the joint space and disposed in the plug opening.
3. The orthopaedic implant system of claim 2, wherein the plug comprises a head defining a head diameter that is greater than the plug opening diameter.
4. The orthopaedic implant system of claim 3, wherein the plug comprises a cylindrical section and a tapered section defining a tapered diameter that decreases in a direction from the cylindrical section towards a bottom of the plug.
5. The orthopaedic implant system of claim 4, wherein the tapered diameter is less than the plug opening diameter.
6. The orthopaedic implant system of claim 3, wherein a portion of the plug extends out of the plug opening when the plug is fully inserted in the plug opening so the head of the plug abuts against the orthopaedic implant, the portion of the plug extending out of the plug opening defining a protruding length of 2-4 mm.
7. The orthopaedic implant system of claim 2, wherein the plug comprises a polymer.
8. The orthopaedic implant system of claim 2, further comprising a graft compressed by the plug.
9. The orthopaedic implant system of claim 8, wherein the graft comprises biological tissue.
10. The orthopaedic implant system of claim 1, comprising one or more additional fixation openings aligned along a longitudinal axis of the orthopaedic implant with the first fixation opening and one or more additional fixation openings aligned along a width axis of the orthopaedic implant with the second fixation opening.
11. The orthopaedic implant system of claim 1, wherein the orthopaedic implant comprises a first region including the first fixation opening and a second region including the second fixation opening, the first region being connected to the second region by an offsetting region so the first region is lower than the second region.
12. The orthopaedic implant system of claim 11, wherein the offsetting region creates a step between the first region and the second region, the step being between 2-4 mm.
13. The orthopaedic implant system of claim 1, wherein the orthopaedic implant defines a greatest width and the diameter of the plug opening is greater than 50% of the greatest width.

14. The orthopaedic implant system of claim 1, wherein the diameter of the plug opening is at least double the diameter of any of the fixation openings.
15. The orthopaedic implant system of claim 1, wherein the orthopaedic implant comprises a second plug opening and wherein a diameter of the second plug opening is greater than a diameter of any fixation openings of the orthopaedic implant.
16. The orthopaedic implant system of claim 15, wherein the second plug opening is configured to accept a second plug inserted in a joint space between the first bone and the second bone or between the second bone and a third bone.
17. The orthopaedic implant system of claim 15, wherein the diameter of the second plug opening is the same as the diameter of the plug opening.
18. An orthopaedic implant system, comprising: an orthopaedic implant comprising: a first fixation opening through the orthopaedic implant; a second fixation opening through the orthopaedic implant; and a plug opening through the orthopaedic implant; a plug configured to be advanced into the plug opening; wherein: a diameter of the plug opening is greater than a diameter of the first fixation opening; the diameter of the plug opening is greater than a diameter of the second fixation opening; the first fixation opening is configured to receive a first fixation element that is configured to be advanced through the first fixation opening and into bone; and the second fixation opening is configured to receive a second fixation element that is configured to be advanced through the second fixation opening and into bone; the plug opening is positioned between the first fixation opening and the second fixation opening and is configured to receive the plug therein; the orthopaedic system is configured such that, in an operable state with the plug fully inserted into and engaged with the plug opening, the plug extends beyond a bottom surface of the orthopaedic implant by a length of 1-7 mm.
19. The orthopaedic implant system of claim 18, wherein the plug is configured to cause compression distraction of a first bone and a second bone when the plug is inserted into a joint space between the first bone and the second bone and disposed in the plug opening.
20. The orthopaedic implant system of claim 18, wherein the plug comprises one or more threads on an outside surface thereof and the plug opening comprises one or more threads on an inside surface thereof configured to engage with the one or more threads of the plug.
21. The orthopaedic implant system of claim 18, wherein the orthopaedic system is configured such that, in the operable state with the plug fully inserted into and engaged with the plug opening, the plug extends beyond the bottom surface of the orthopaedic implant by a length of 2-4 mm.
22. An orthopaedic implant system, comprising: an orthopaedic implant comprising: a first fixation opening through a first portion of the orthopaedic implant; a second fixation opening through a second portion of the orthopaedic implant; and a plug opening through the orthopaedic implant; and a step configured such that a bottom surface of the first portion of the orthopaedic implant is offset from a bottom surface of the second portion of the orthopaedic implant; a plug configured to be advanced into the plug opening; wherein: a diameter of the plug opening is greater than a diameter of the first fixation opening; the diameter of the plug opening is greater than a diameter of the second fixation opening; the first fixation opening is configured to receive a first fixation element that is configured to be advanced through the first fixation opening in an operable state of the orthopaedic implant system; and the second fixation opening is configured to receive a second fixation element that is configured to be advanced through the second fixation opening in an operable state of the orthopaedic implant system.
23. The orthopaedic implant system of claim 22, wherein the step is positioned between the first portion of the orthopaedic implant and the second portion of the orthopaedic implant.